

Educations

Ph.D. in Electrical and Electronic Engineering Aug 2026 - present
Supervised by [Prof. Lihua Xie](#) Nanyang Technological University ([NTU](#)), Singapore
Master Degree in Control Engineering and Automation Jul 2024 - Jul 2026
Supervised by [Prof. Tung Lam Nguyen](#) Hanoi Univ of Science and Technology ([HUST](#)), Hanoi, Vietnam
Thesis: Design control structures for parallel platforms in maritime applications
Bachelor Degree in Control Engineering and Automation Oct 2020 - Apr 2024
Supervised by [Prof. Tung Lam Nguyen](#) Hanoi Univ of Science and Technology ([HUST](#)), Hanoi, Vietnam
Thesis: Balancing, Motion Planning, and Tracking Control for Ballbot Systems

Work experiences

R&D Robotics Engineer at VinRobotics, Vingroup Sep 2025 - Jul 2026
Full-time, supervised by [Dr. Minh Nhat Vu](#) Hanoi, Vietnam
Advanced control, impedance control, motion planning, optimization, obstacle avoidance, manipulator, humanoid, conduct scientific paper, etc.
Research Project Assistant at HUST Oct 2021 - Dec 2026
Contract (Hybrid), supervised by [PhD. Chinh Hoang Duc](#) and [Assoc.Prof.PhD. Tung Lam Nguyen](#) Hanoi, Vietnam
Safety control, Observer, Nonlinear control, Motion control/planning, ball-balancing robot, parallel mechanism, underwater vehicles, vibration compensation, etc.

Selected publications

- [7] **Vu, D. C.**, Nguyen, V. T., Nguyen, D. H., Nguyen, M. C., Tran, V. T., & Vu, M. N. (2026). Real-Time Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators. *IEEE RA-L*. [\[pdf\]](#)
- [6] Nguyen, T. T. H., **Vu, D. C.**, Pham, M. D., Nguyen, T. L., & Nguyen, T. V. A. (2026). Unifying hierarchical sliding mode control and control barrier function for tilt angle constraint of a ball-balancing robot. *IET Cyber-Systems and Robotics*. [\[pdf\]](#)
- [5] **Vu, D. C.**[†], Tran, S.[†], Nguyen, T. L., & Hoang, D. C. (2025). Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks. *Ocean Engineering*. [\[pdf\]](#) [†]equal contribution
- [4] **Vu, D. C.**, Nguyen, D. H., Vu, M. N., & Nguyen, T. L. (2025). Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations. *Ocean Engineering*, 340, 122142. [\[pdf\]](#)
- [3] Pham, M. D., **Vu, D. C.**, Nguyen, T. T. H., Nguyen, T. V. A., Vu, M. N., & Nguyen, T. L. (2025). CBFs-based model predictive control for obstacle avoidance with tilt angle limitation for ball-balancing robots. *IEEE Access*. [\[pdf\]](#)
- [2] **Vu, D. C.**, Nguyen, D. H., & Nguyen, T. L. (2024). A novel approach of consensus-based finite-time distributed sliding mode control for Stewart platform manipulators motion tracking. *Results in Engineering*. [\[pdf\]](#)
- [1] **Vu, D. C.**, Pham, M. D., Nguyen, T. T. H., Nguyen, T. V. A., & Nguyen, T. L. (2024). Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints. *International Journal of Robust and Nonlinear Control*. [\[pdf\]](#)

Honours & awards

NTU Research Scholarship Nanyang Technological University (NTU), Singapore
Master, PhD Scholarship Programme Vingroup Innovation Foundation (VINIF)
Best Thesis Defense Award Hanoi University of Science and Technology

Academic activities

Invited review for *Nonlinear Dynamics, Ocean Engineering, IEEE Access, IEEE Robotics and Automation Letters (RA-L); International Conference on Intelligent Robots and Systems (IROS 2026), IEEE RO-MAN 2026, Humanoids 2026, International Conference on Robotics and Automation (ICRA 2027).*